

CS 663: Digital Image Processing

Assignment 1: Problem 1(f) Report

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1 Image Subsampling and Interpolation

1.1 1(f) Image Downsampling and Upsampling on Chest CT Scans

Theory and Implementation

The objective of this problem is to reconstruct a full-resolution chest CT scan from a subsampled version using three different interpolation techniques. The provided data contains two real-valued 2D images, one original and one subsampled, originally stored in a MATLAB `.mat` format. To process this in a native C++ environment, the data was exported to CSV format, ingested into flat 1D `std::vector<float>` arrays, and mathematically scaled back to the original high-resolution dimensions.

A critical strict geometric constraint for this implementation was ensuring the outer-most boundaries aligned perfectly: the first and last rows, as well as the first and last columns of the subsampled image, must map exactly to their respective positions in the enlarged image. This was achieved by using a proportional coordinate mapping function that scales the output index by the ratio of the maximum dimension bounds, ensuring the spatial geometry of the chest cavity is not warped or shifted.

The three interpolation algorithms applied are:

- **Nearest-Neighbor:** Maps the target coordinate and rounds to the closest integer source index, directly copying the intensity value.
- **Bilinear:** Computes a linear weighted average across a 2×2 pixel neighborhood based on sub-pixel fractional distances.
- **Bicubic:** Evaluates a 4×4 pixel neighborhood using a piecewise cubic spline convolution to fit a continuous polynomial surface over the data points.

Observations and Difference Analysis

To evaluate the quality of the interpolation schemes, the original, reconstructed, and difference images (Original - Enlarged) are visualized using the “jet” colormap with strictly uniform colormap limits applied across all respective figures.

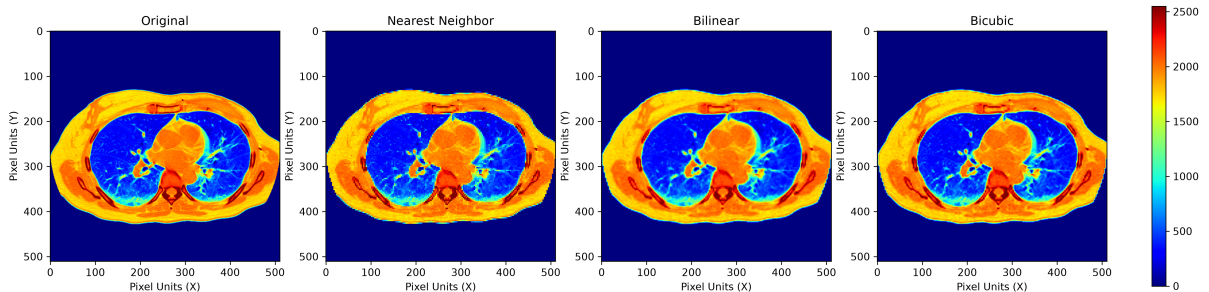


Figure 1: Comparison of the original CT scan and the three enlarged outputs reconstructed from the subsampled data. Displayed using the “jet” colormap.

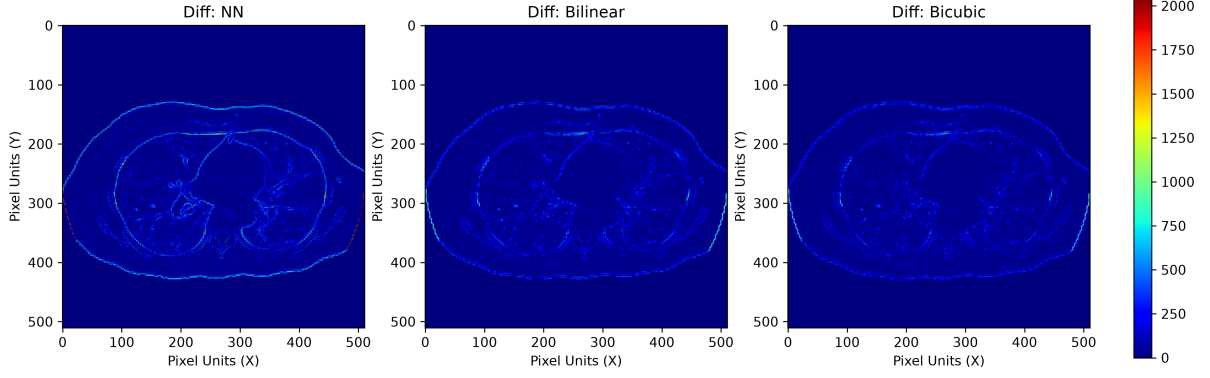


Figure 2: Difference images (Original - Enlarged) highlighting the spatial error introduced by each interpolation method. Higher intensities (warmer colors) indicate greater error.

Visual inspection of the difference images reveals the distribution of the reconstruction error. The nearest-neighbor difference image contains severe high-magnitude residuals (visible as bright red/yellow outlines) concentrated precisely around the high-frequency edges of the chest cavity and internal lung structures. Bilinear interpolation significantly reduces these severe edge artifacts by introducing gradients, but still exhibits generalized mid-level error across textured anatomical regions. The bicubic difference image is the darkest overall (predominantly deep blue), indicating that its residuals are uniformly the lowest across both sharp edges and smooth tissue gradients.

Quantitative Error Analysis (RMSE)

The visual inferences are strictly confirmed by computing the Root Mean Square Error (RMSE) with respect to the original image acting as the ground truth.

Interpolation Method	RMSE
Nearest-Neighbor	103.4262
Bilinear	68.9653
Bicubic	63.3156

Table 1: Root Mean Square Error (RMSE) for the three interpolation schemes.

Why Bicubic Yields the Minimum RMSE:

Bicubic interpolation achieves the lowest RMSE because it approximates the continuous nature of underlying anatomical tissues far better than the alternatives. While nearest-neighbor assumes data acts as flat step-functions (causing massive error at boundaries) and bilinear assumes strictly linear planes between points, bicubic interpolation uses a 3rd-order polynomial to model smooth curves. By evaluating 16 surrounding pixels (a 4×4 grid), it accounts for both the rate of change (first derivative) and the curvature (second derivative) of the image intensities. This higher-order curve fitting successfully preserves the sharpness of structural boundaries in the CT scan without introducing the severe blocky artifacts of nearest-neighbor or the excessive blurring inherent to bilinear averaging, thereby mathematically minimizing the residual differences from the ground-truth high-resolution data.